MIL-D-45477B (MU)
9 January 1970
SUPERSEDING
MIL-D-45477A
17 June 1965

MILITARY SPECIFICATION

DETONATORS, PERCUSSION, DELAY, 15 SECOND, M1A2 AND 8 SECOND, M2A1, PARTS FOR

1. SCOPE

1.1 This specification covers the parts for one type of detonator, delay designated as Detonator, Percussion, Delay, 15 Second, M1A2 and 8 Second, M2A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC: 1375

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:07Z
Check the source to verify that this is the current version before use.

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8799905 - Cap Protector Assembly
8799915 - Housing, Delay
8799917 - Holder, Primer
8799918 - Firing Pin Assembly

PUBLICATIONS

U.S. ARMY MUNITIONS COMMAND

ET-8816147 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on drawings (dwg.) 8799905, 8799915, 8799917 and 8799918, and with all requirements specified in applicable specifications.

3.3 Embrittlement of springs.-The firing pin spring shall not break when extended to the point of minimum permanent distortion, when tested as specified in 4.4.2.

3.4 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.5 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

Source: https://assist.dla.mil - Downloaded: 2026-10-22T03:07Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-Prior to the start of regular production, the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:07Z
Check the source to verify that this is the current version before use.

officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 25 complete sets of parts and 25 subassemblies (i.e., 25 each of every component and every subassembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect production uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Components and subassemblies will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any component or subassembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any component or subassembly in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:07Z
Check the source to verify that this is the current version before use.

applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and the same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Spring, firing pin (see dwg. 8799924 covering a detail of dwg. 8799918)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:07Z
Check the source to verify that this is the current version before use.

Major:	AQL 0.65 Percent		
101.	Load test failure at assembled height	Gage	01001
102.	Load at .90 inch	Gage	01008
103.	Protective finish missing or damaged	Visual	01005
Minor:	AQL 1.50 Percent		
201.	Inside diameter, min.	Gage	01003
202.	Outside diameter, max.	Gage	01004
203.	Ends not closed and ground	Visual	01002
204.	Evidence of poor workmanship (see 3.4)	Visual	01006

4.3.2.2 Firing pin (see dwg. 8799921 covering a detail of dwg. 8799918)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Large outside diameter	Gage	02001
102.	Diameter of shank, maximum (max.)	Gage	02002
103.	Location of cotter pin hole	Gage	02003
104.	Diameter of cotter pin hole, minimum (min.)	Gage	02004
105.	Profile of point	Gage	02005
106.	Protective finish missing or damaged	Visual	02006
Minor:	AQL 1.50 Percent		
201.	Total length	Gage	02007
202.	Finish improper in cotter pin hole	Visual	02008
203.	Evidence of poor workmanship (see 3.4)	Visual	02009

4.3.2.3 Housing, firing pin (see dwg. 8799920 covering a detail of dwg. 8799918)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:07Z
Check the source to verify that this is the current version before use.

Major:	AQL 1.00 Percent		
101.	Inside diameter, min.	Gage	03001
102.	Diameter of cotter pin hole, min.	Gage	03002
103.	Location of cotter pin hole	Gage	03003
104.	Pitch diameter of thread, max	Gage	03004
105.	Minor diameter of thread, max.	Gage	03005
106.	True position of firing pin hole with inside diameter	Gage	03007
Minor:	AQL 4.00 Percent		
201.	Diameter of firing pin hole, min.	Gage	03006
202.	Diameter of counterbore, min.	Gage	03008
203.	Total length	Gage	03009
204.	Depth of counterbore, min.	Gage	03010
205.	Distance to end of threads, min.	Gage	03011
206.	Hole by thread missing	Visual	03012
207.	Protective finish missing or damaged	Visual	03013
208.	Evidence of poor workmanship (see 3.4)	Visual	03014

4.3.2.4 Firing pin assembly (see dwg. 8799918)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Cotter pin not spread	Visual	04001
Major:	AQL 0.40 Percent		
101.	Component missing	Visual	04003

Minor: None defined

4.3.2.5 Primer holder (see dwg. 8799917)

Categories	Defects	Method of Inspection	Code No.

Critical: None defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:07Z
Check the source to verify that this is the current version before use.

Major:	AQL 0.65 Percent		
101.	Diameter of flash hole	Gage	05001
102.	Diameter of primer cavity	Gage	05002
103.	Depth of primer cavity	Gage	05003
104.	Outside diameter at datum	Gage	05004
Minor:	AQL 1.50 Percent		
201.	Total length	Gage	05005
202.	Corners at rear end not flattened	Visual	05006
203.	Evidence of poor workmanship, (see 3.4)	Visual	05007

4.3.2.6 Housing, delay (see dwg. 8799915)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Location of large datum diameter	Gage	06001
102.	Location of small datum diameter	Gage	06015
103.	Pitch diameter of firing pin housing threads, min.	Gage	06002
104.	Pitch diameter of protector cap threads, min.	Gage	06003
105.	Major diameter of firing pin housing threads, min.	Gage	06016
106.	Major diameter of protector cap threads, min.	Gage	06017
107.	Diameter of delay charge cavity	Gage	06004
108.	Depth of delay charge cavity	Gage	06005
109.	Flash hole missing	Visual	06006
Minor:	AQL 2.50 Percent		
201.	Length of firing pin housing threads, min.	Gage	06007
202.	Total length	Gage	06008
203.	Length of blasting cap crimping surface	Gage	06009
204.	Diameter crimping surface, max.	Gage	06010
205.	Crimping grooves missing	Visual	06011

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:07Z
Check the source to verify that this is the current version before use.

206.	Knurl missing	Visual	06012
207.	Evidence of poor workmanship (see 3.4)	Visual	06014

4.3.2.7 Cap protector assembly (see dwg. 8799905)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 4.00 Percent		
201.	Pitch diameter of thread, max.	Gage	07001
202.	Minor diameter of thread, max.	Gage	07009
203.	Inside diameter, min.	Gage	07002
204.	Length of thread, min.	Gage	07003
205.	Total length, max.	Gage	07005
206.	Thickness through bottom, max.	Gage	07006
207.	Ferrule missing or loose	Visual	07007
208.	Evidence of poor workmanship (see 3.4)	Visual	07008

4.3.3 Testing

4.3.3.1 Cotter safety and split pin withdrawal at lower limit (see dwg. 8799918), Major Defect, Code No. 04001.-

4.3.3.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, the sampling plan for this test shall be in accordance with MIL-STD-1235, CSP-1 and an AQL of 0.15 percent, wherein i = 740 and F = 1/100. The test shall be performed as specified in 4.4.1 using equipment as specified in 4.3.4.

4.3.3.1.2 After three consecutive lots have met the criteria of 4.3.3.1.1, the sampling plan for this test shall be in accordance with MIL-STD-1235, CSP-1 and an AQL of 0.25 percent, wherein i = 450 and F = 1/100. The test shall by performed as specified in 4.4.1 using equipment specified in 4.3.4.

4.3.3.2 Cotter safety and split pin withdrawal at upper limit (see dwg. 8799918), Major Defect, Code No. 04005.-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:07Z
Check the source to verify that this is the current version before use.

4.3.3.2.1 Beginning with the first lot produced and continuir until three consecutive lots have complied with the acceptance criteria specified, a single sampling plan shall be used for this test in accordance with MIL-STD-105 using code letter M and an AQL of 2.50 percent. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.2.2 After three consecutive lots have met the criteria of 4.3.3.2.1, a single sampling plan shall be used in accordance with MIL-STD-105 using code letter K and an AQL of 2.50 percent. This test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.3 Spring embrittlement (See 3.3), Major Defect, Code No. 01007.-A sample of 30 firing pin springs shall be selected from each lot for this test. If any spring fails to comply with the requirements of 3.3, the lot shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.4 Inspection equipment.-Equipment Tabulation Number ET-8816147 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Cotter safety and split pin withdrawal.-The firing pin assembly shall be secured in a horizontal position with the "eyes" of the cotter pins pointed directly down. A weight, equal to the minimum withdrawal force specified on the drawing, shall be attached to each cotter pin in such a manner that the force applied to the cotter pin does not exceed that specified. This weight shall be replaced, in the manner specified above, by a weight equal to the maximum withdrawal force. An assembly will be classed defective if either or both cotter pins are withdrawn by the first weight or if either or both cotter pins fail to be withdrawn by the second weight.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:07Z
Check the source to verify that this is the current version before use.

4.4.2 Spring embrittlement.-The spring shall be repeatedly stretched with increasing loads until it does not return to its original free height. That point shall indicate minimum distortion. Any unit which fractures prior to extension to minimum distortion shall be classed defective. Springs so tested shall be scrapped.

5. PREPARATION FOR DELIVERY

5.1 Level C.-Inert component parts shall be packaged, packed and marked in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as referenced on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicated the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:07Z
Check the source to verify that this is the current version before use.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicatted by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:07Z
Check the source to verify that this is the current version before use.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1375-A-059

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:07Z
Check the source to verify that this is the current version before use.